

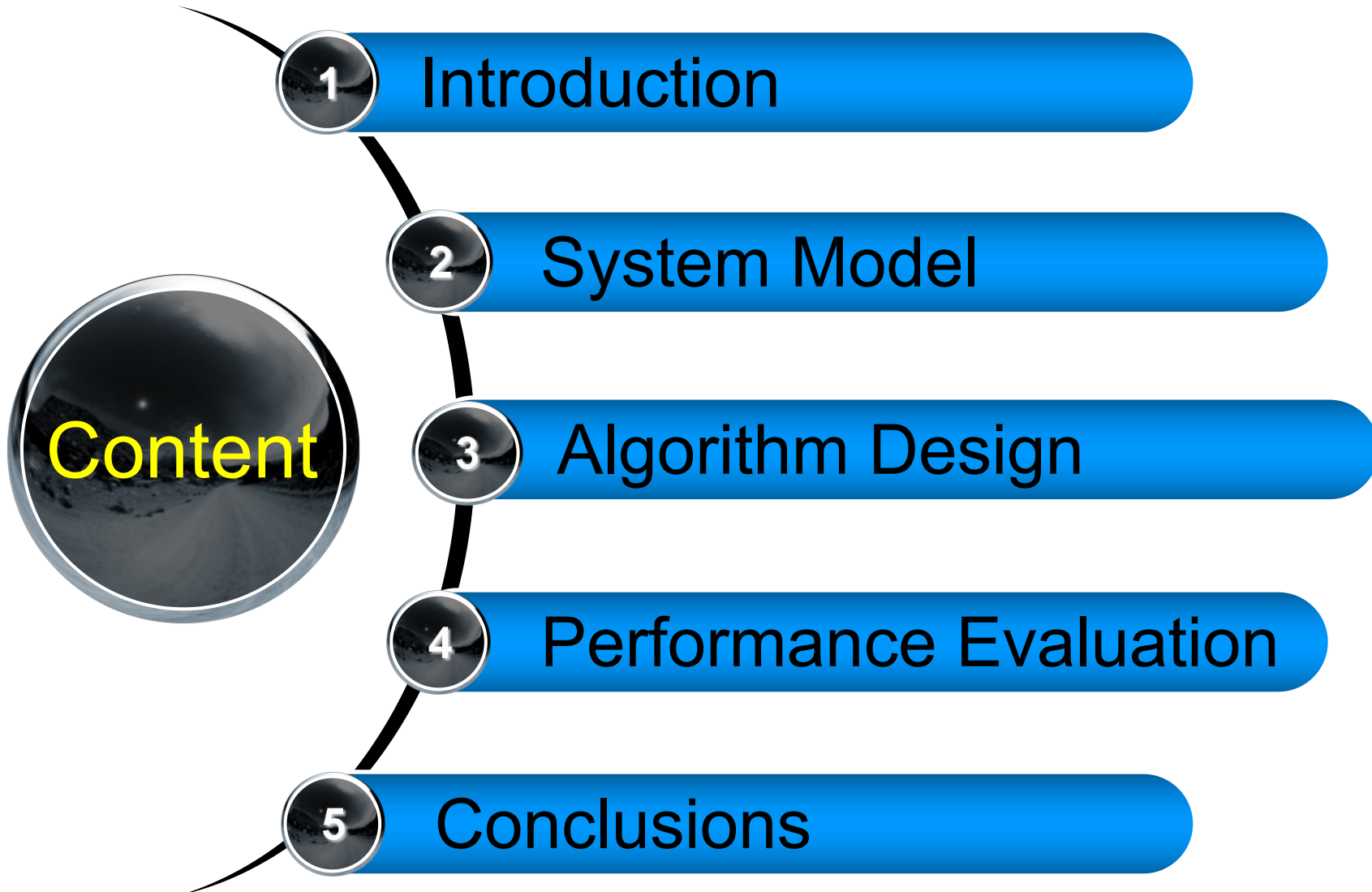
Efficient LLM Edge Collaboration Deployment with LoRA

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Introduction

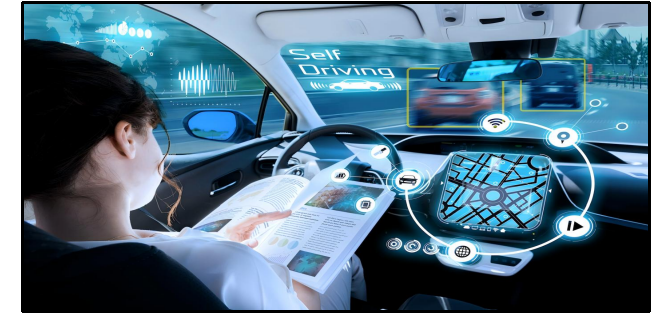
- LLMs demonstrate substantial potential across many application domains.



NLP

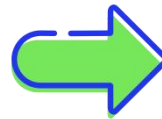
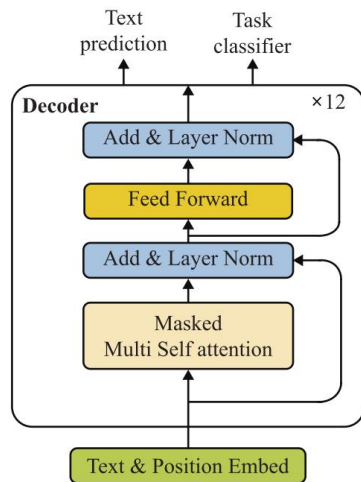


Healthcare

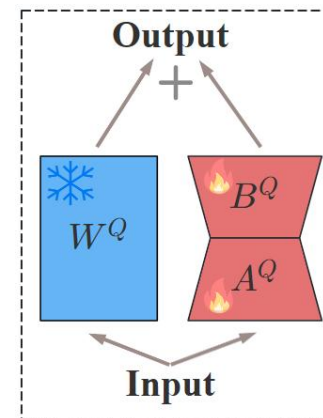


Autonomous Driving

- Common LLMs are built on the Transformer architecture.



Base Models



LoRA



Personalized LLMs

Introduction

- LLM in Cloud Data Center

- ✓ Abundant computing and storage resources
- ✗ High latency and network jitter
- ✗ User data security and privacy issues

Unsuitable for latency-sensitive applications

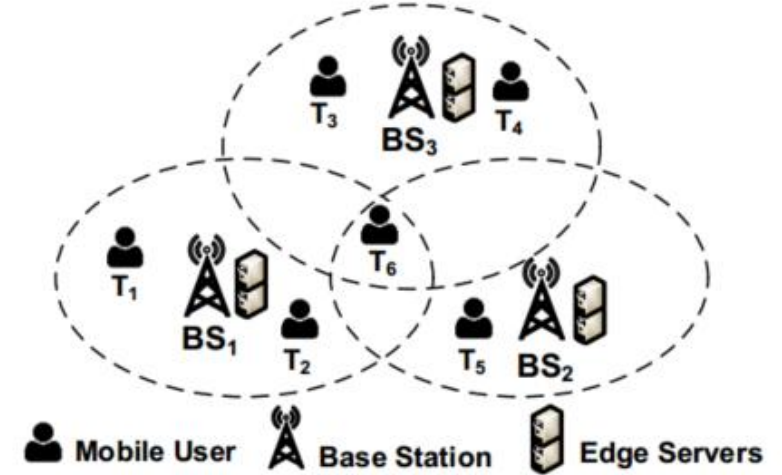
- LLM in Mobile Devices

- ✓ User privacy is guaranteed
- ✓ Low latency and stable inference
- ✗ Computing capacity, storage capacity, and battery capacity are extremely limited

Difficult to support high-performance LLMs

Introduction

- **Edge computing** is an attractive paradigm to solve the problem mentioned above.
 - Edge servers are located closer to users than cloud data centers, resulting in lower inference latency.
 - Edge servers have greater computing and storage capabilities than mobile devices.



However, the number of LLMs makes it impossible for edge servers to cache them all.

Question: How to deploy LLMs on edge servers efficiently?



LLM deployment problem

Introduction

- Challenges of LLM deployment in MEC

Challenge 1: Abundant LLMs **compete for limited storage capacity** of edge servers.



Challenge 2: Conventional deployment approaches may result in **redundant caching**.



Challenge 3: **Reduce deployment costs** while ensuring **low-latency service quality**.

Introduction

- Related work

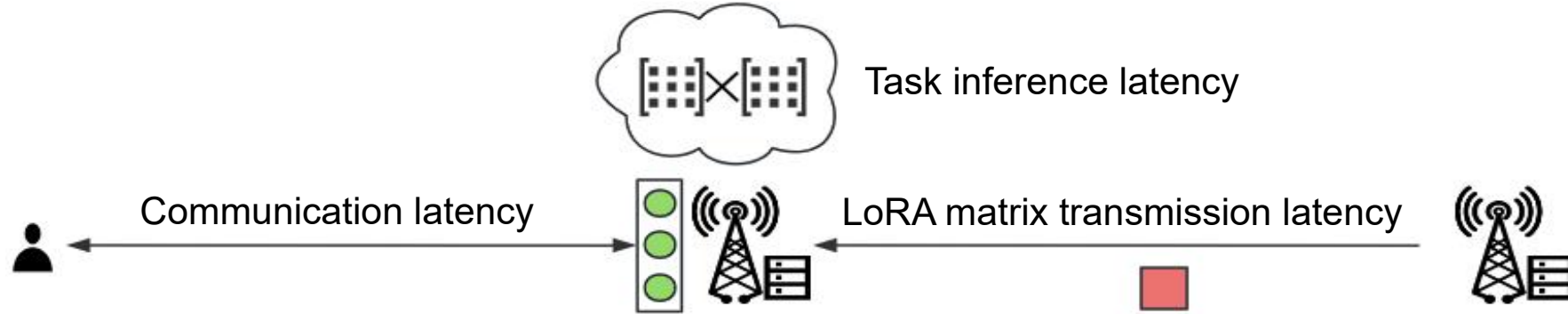
	Deployment cost	Consider redundant cache
Edgeshare[IOTJ'24]	×	×
DDPG[WCNC'25]	×	×
LC[GLOBECOM'23]	×	×
Trimcaching[ICDCS'24]	×	✓
EdgeColl	✓	✓

System Model

- **Edge server:**
 - Coverage area
 - Limited storage capacity
- **User:**
 - Send an inference request to a specific LLM
- **LLM:**
 - Consist of a base model and a LoRA matrix
- **Decision variables:**
 - Base model deployment decision
 - LoRA matrix deployment decision
 - User request routing decision

System Model

- Total inference latency



- Objective: Minimize average inference latency and LLM deployment costs

$$\text{minimize} \quad \underbrace{\alpha \frac{\sum_{u_i \in U} T_{i,j}}{|U|}}_{\text{User average inference latency}} + (1 - \alpha) \underbrace{\left(\sum_{s_j \in S} \sum_{f_l \in F} c_{j,l} y_{j,l} + \sum_{s_j \in S} \sum_{r_h \in R} c_{j,h} x_{j,h} \right)}_{\text{LLM deployment cost}}$$

$$\text{subject to} \quad \sum_{f_l \in F} D_l^{\text{Base}} y_{j,l} + \sum_{r_h \in R} D_h^{\text{LoRA}} x_{j,h} \leq q_j, \quad \forall s_j \in S, \quad \longrightarrow \text{Capacity constraint}$$

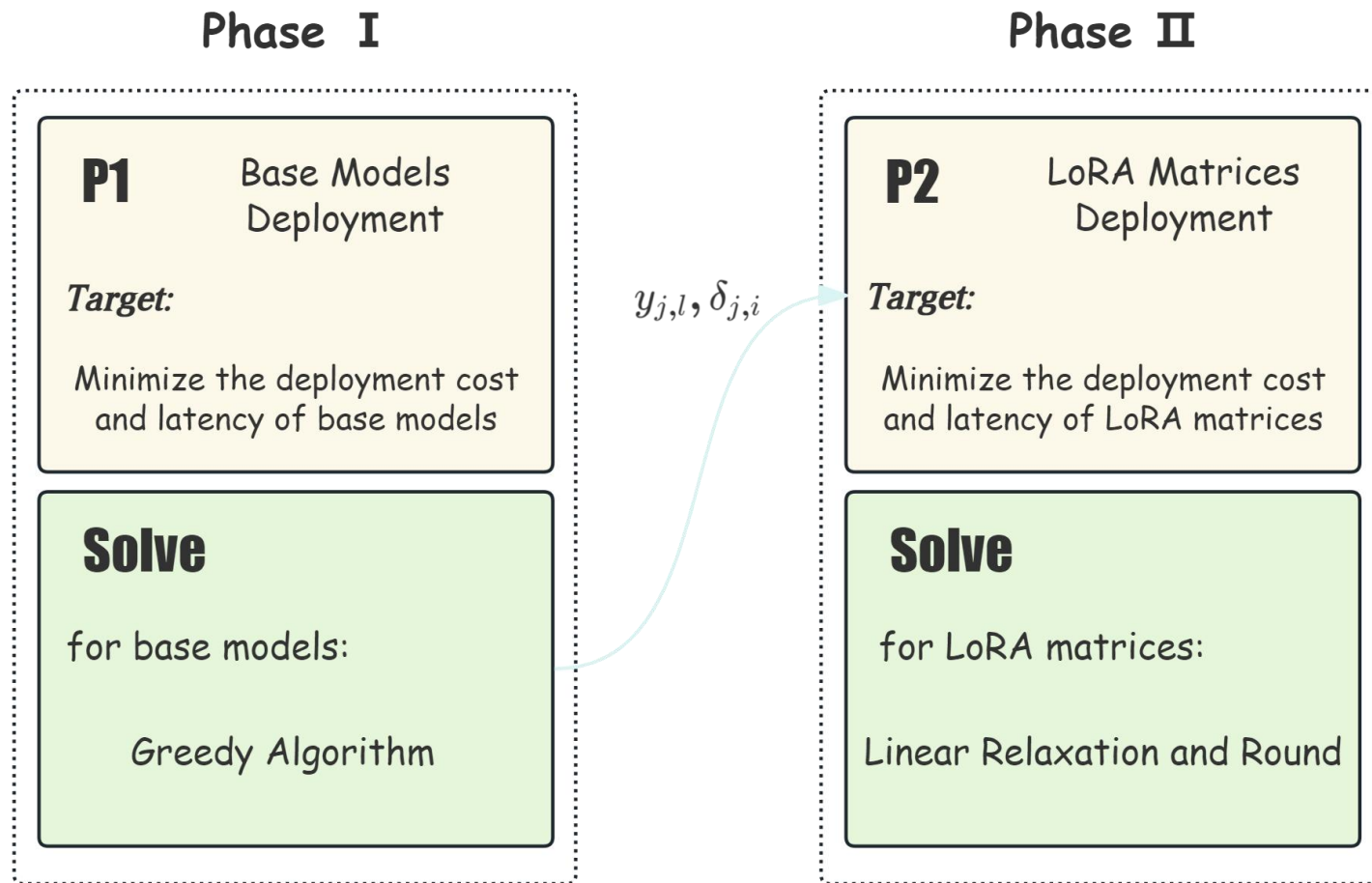
$$\delta_{j,i} = 0, \quad \forall u_i \notin \text{cov}(s_j), \quad \longrightarrow \text{Coverage constraint}$$

$$\delta_{j,i} \leq y_{j,l}, \quad \forall u_i \in U, \forall s_j \in S, \quad \longrightarrow \text{Routing constraint}$$

$$y_{j,l} \in \{0,1\}, x_{j,h} \in \{0,1\}, \delta_{j,i} \in \{0,1\}, \quad \forall u_i \in U, \forall s_j \in S.$$

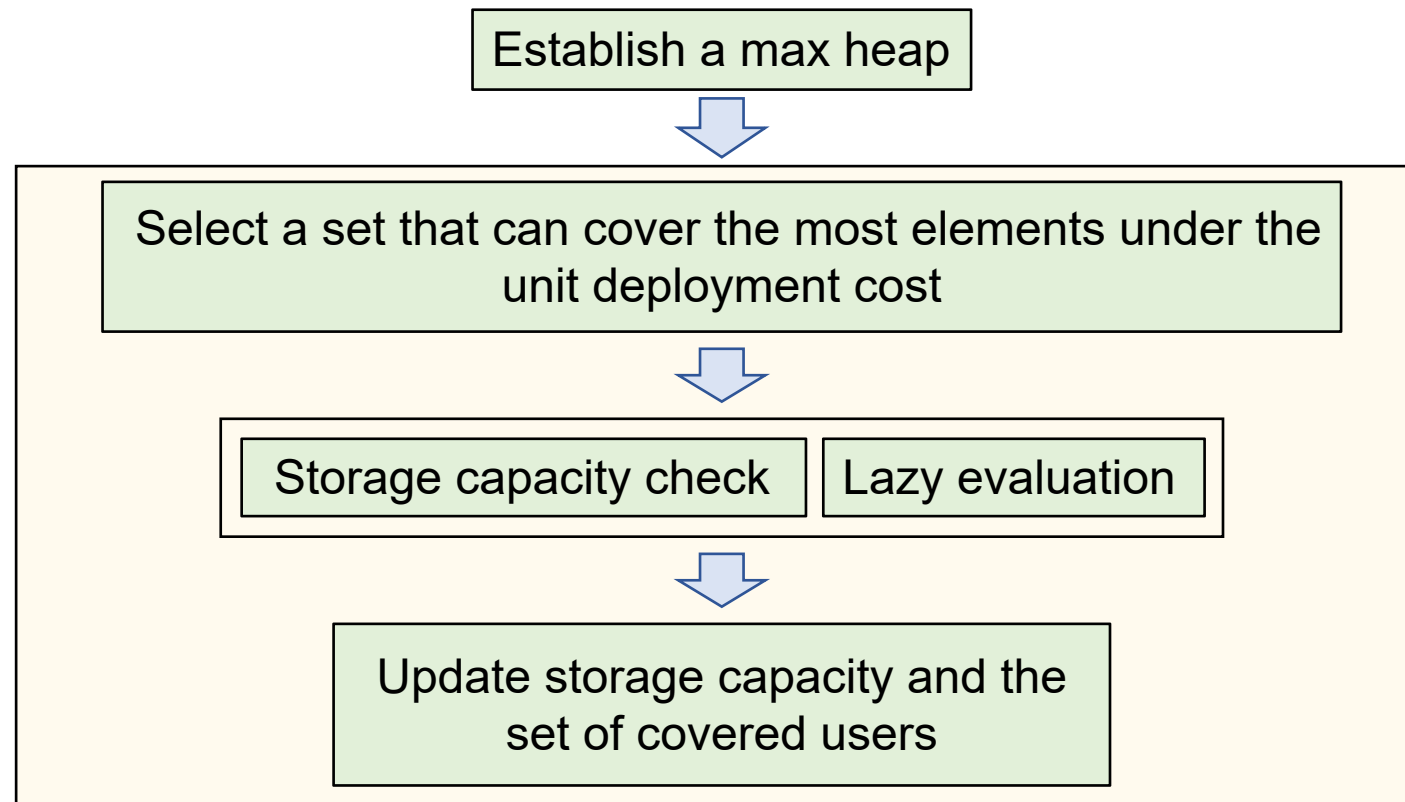
Algorithm Design: *EdgeColl*

- *EdgeColl* decomposes the LLM deployment problem into **two related subproblems** solved in sequence.



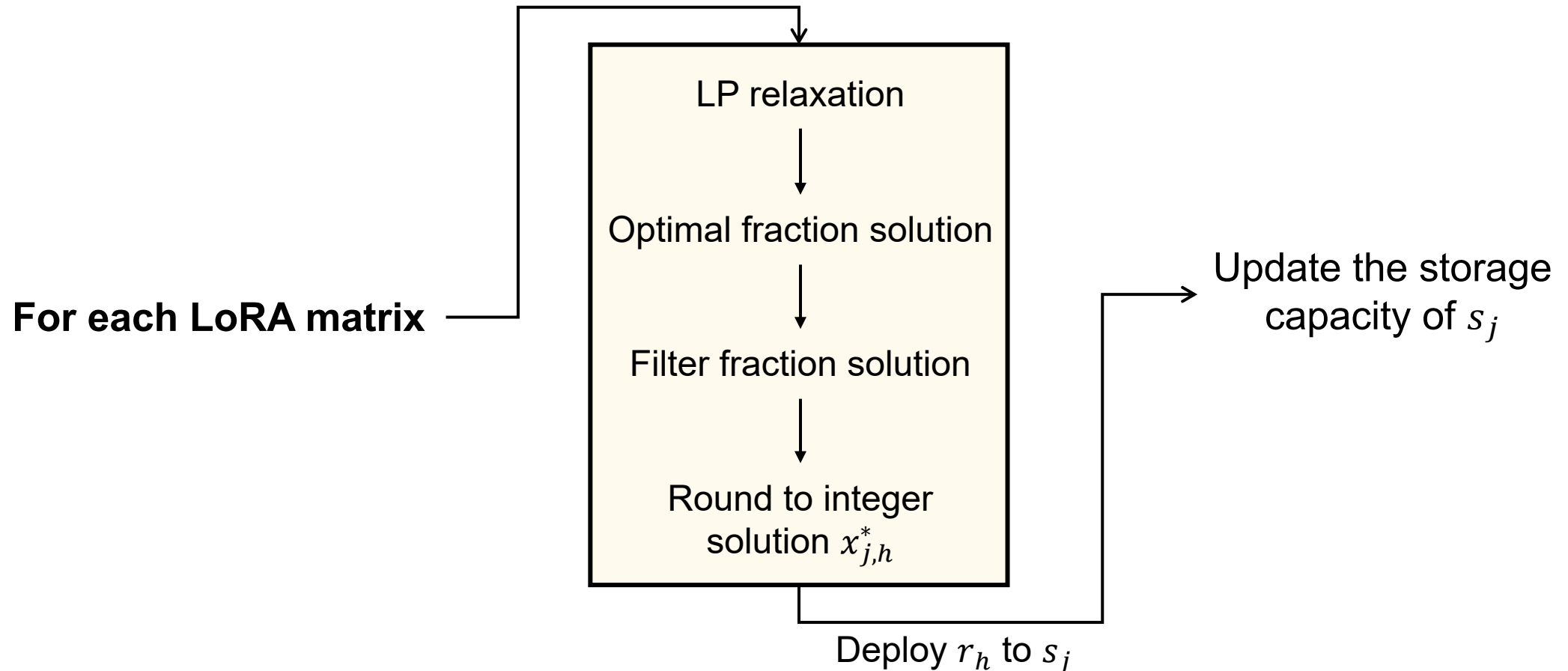
Algorithm Design: *EdgeColl*

- Solve *base models deployment* problem
 - (user, base model) --> an element
 - (edge server, base model) --> a set



Algorithm Design: *EdgeColl*

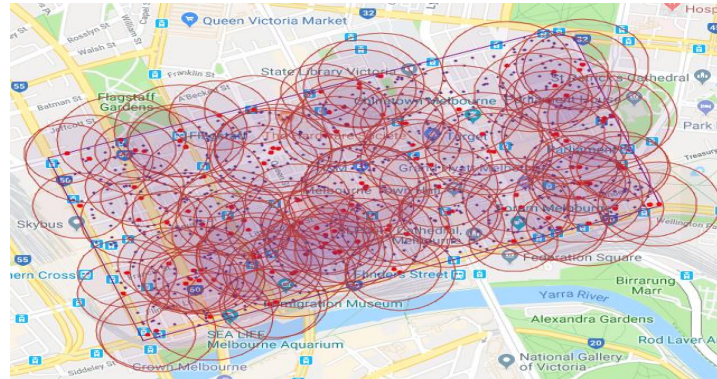
- Solve *LoRA matrices deployment* problem
 - Known base models deployment results y^*



Performance Evaluation

Experiment settings

- EUA Dataset:
 - The central business district, Melbourne, Australia
 - 125 edge servers, 816 mobile users



- Parameters
 - Three base models: Llama2-70B, Qwen2.5-72B and PaLM-62B
 - Number of mobile users: from 200 to 800
 - Number of edge servers: from 50 to 110
 - Storage capacity of edge server: from 60GB to 660GB

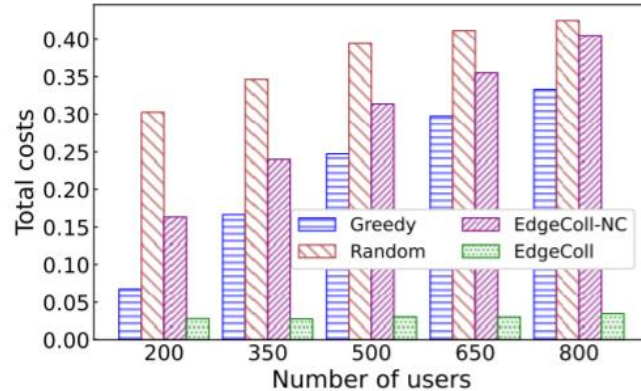
Performance Evaluation

Experiment settings

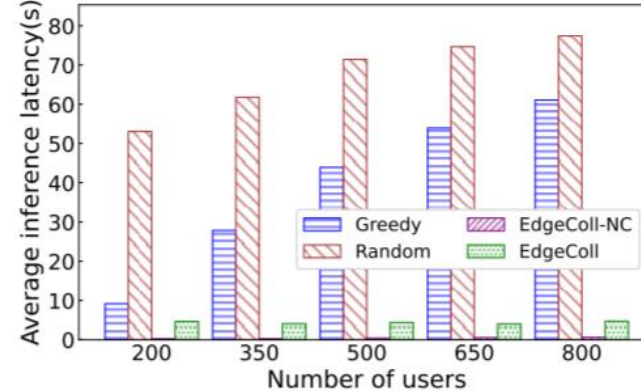
- Three benchmarks:
 - **Greedy:** Deploy the entire LLM greedily on each edge server to minimize inference latency.
 - **Random:** Randomly deploy the entire LLM on each edge server.
 - **EdgeColl-NC:** Under the EdgeColl framework, minimize the inference latency while ignoring LLM deployment costs.
- Performance metric:
 - Total costs
 - Average inference latency (s)
 - Deployment costs

Performance Evaluation

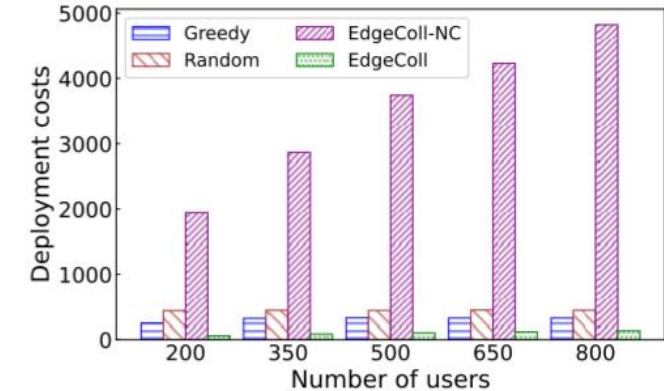
- Impact of mobile users



(a) Total costs vs. # of mobile users



(b) Inference latency vs. # of mobile users

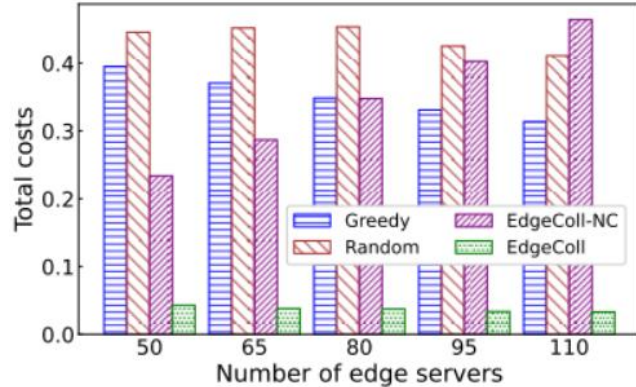


(c) Deployment costs vs. # of mobile users

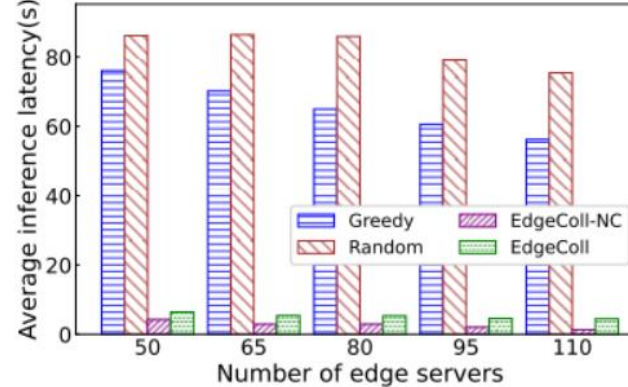
Results: With more users, the total costs of all approaches rise, while **EdgeColl** consistently maintains the lowest total costs, reducing the average total costs by **more than 80%** compared with benchmarks.

Performance Evaluation

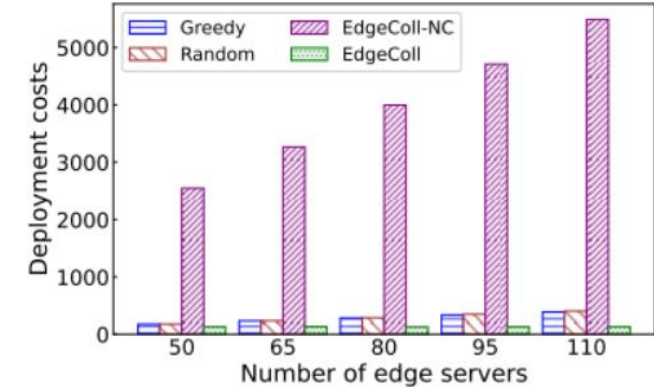
- Impact of edge servers



(a) Total costs vs. # of edge servers



(b) Inference latency vs. # of edge servers

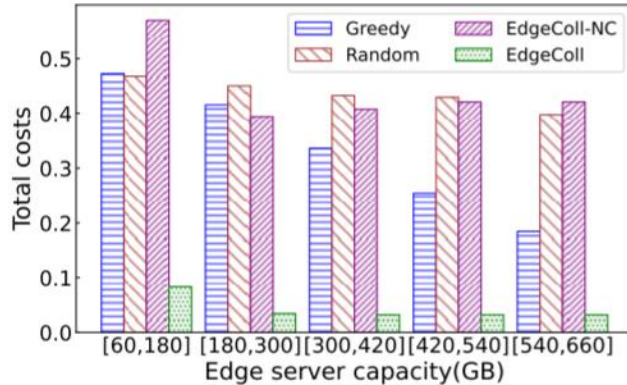


(c) Deployment costs vs. # of edge servers

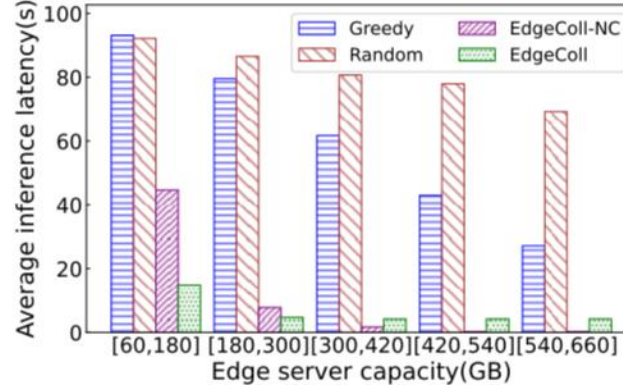
Results: As the number of edge servers increases, **EdgeColl** achieves continuous reductions in total costs and inference latency, while maintaining nearly stable deployment costs, demonstrating its excellent scalability.

Performance Evaluation

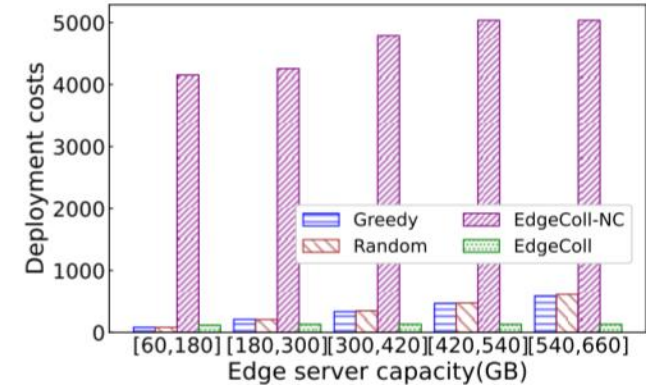
- Impact of storage capacity



(a) Total costs vs. Storage capacity



(b) Inference latency vs. Storage capacity



(c) Deployment costs vs. Storage capacity

Results: With larger storage capacity,

- (1) Both the total costs and inference latency of all approaches decrease.
- (2) **EdgeColl** still significantly outperforms other approaches by making more efficient use of storage resources for model deployment.

Conclusions

- A novel LLM deployment architecture that integrates base models and LoRA matrices to achieve personalized inference.
- We present ***EdgeColl*** to jointly optimize the inference latency and LLM deployment costs.
- The simulation results show the performance of ***EdgeColl***.



Thank you!

Q&A